

Frameworks for Valuation

In Part One, we built a conceptual framework to show what drives the creation of value for investors. A company's value stems from its ability to earn a healthy return on invested capital (ROIC) and its ability to grow. Healthy rates of return and growth produce future cash flows, the ultimate source of value.

Part Two offers a step-by-step guide for analyzing and valuing a company in practice, including technical details for properly measuring and interpreting the drivers of value. Among the many ways to value a company (see Exhibit 10.1 for an overview), we focus particularly on two: enterprise discounted cash flow (DCF) and discounted economic profit. When applied correctly, both valuation methods yield the same results; however, each model has certain benefits in practice. Enterprise DCF remains a favorite of practitioners and academics because it relies on the flow of cash in and out of the company, rather than on accounting-based earnings. For its part, the discounted economic-profit valuation model can be quite insightful because of its close link to economic theory and competitive strategy. Economic profit highlights whether a company is earning its cost of capital and quantifies the amount of value created each year. Given that the two methods yield identical results and have different but complementary benefits, we recommend creating *both* enterprise DCF and economic-profit models when valuing a company.

Both the enterprise DCF and economic-profit models rely on the weighted average cost of capital (WACC). WACC-based models work best when a company maintains a relatively stable debt-to-value ratio. If a company's debt-to-value ratio is expected to change, WACC-based models can still yield accurate results but are more difficult to implement correctly. In such cases, we recommend an alternative to WACC-based models: adjusted present value (APV). APV discounts the same free cash flows as the enterprise DCF model but uses the unlevered cost of equity as the discount rate (without the tax benefit of debt).

EXHIBIT 10.1 Frameworks for DCF-Based Valuation

Model	Measure	Discount factor	Assessment
Enterprise discounted cash flow	Free cash flow	Weighted average cost of capital	Works best for projects, business units, and companies that manage their capital structure to a target level.
Discounted economic profit	Economic profit	Weighted average cost of capital	Explicitly highlights when a company creates value.
Adjusted present value	Free cash flow	Unlevered cost of equity	Incorporates changing capital structure more easily than WACC-based models.
Capital cash flow	Capital cash flow	Unlevered cost of equity	Combines free cash flow and the interest tax shield in one number, making it difficult to compare operating performance among companies and over time.
Equity cash flow	Cash flow to equity	Levered cost of equity	Difficult to implement correctly because capital structure is embedded within the cash flow. Best used when valuing financial institutions.

It then values the tax benefits associated with debt and adds them to the all-equity value to determine the total enterprise value.¹ When applied properly, the APV model results in the same value as the enterprise DCF value.

This chapter also includes a brief discussion of capital cash flow and equity cash flow valuation models. Properly implemented, these models will yield the same results as enterprise DCF. However, given that they mix operating performance and capital structure in cash flow, we believe implementation errors occur more easily. For this reason, we avoid capital cash flow and equity cash flow valuation models, except when valuing banks and other financial institutions, where capital structure is an inextricable part of operations (for how to value banks, see Chapter 38).

ENTERPRISE DISCOUNTED CASH FLOW MODEL

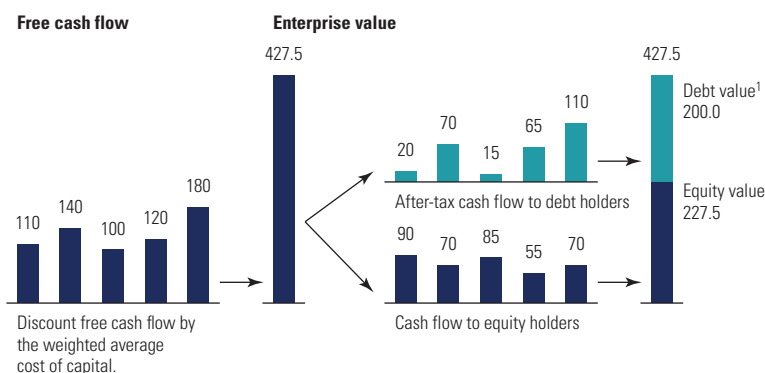
The enterprise DCF model discounts free cash flow (FCF), meaning the cash flow available to all investors—equity holders, debt holders, and any other investors—at the weighted average cost of capital, meaning the blended cost of capital for all investor capital. The company's debt and other nonequity claims on cash flow are subtracted from enterprise value to determine equity value.² Equity valuation models, in contrast, value directly the equity holders' cash flows. Exhibit 10.2 demonstrates the relationship between enterprise value and equity value. For this example, it is possible to calculate equity holders'

¹ Leveraged buyouts conducted by private-equity companies often use substantial leverage to finance the acquisition. In these situations, discount free cash flow at the unlevered cost of equity and evaluate the benefits of financial structure separately.

² Throughout this chapter, we refer to debt and other nonequity claims. Other nonequity claims arise when stakeholders other than shareholders have a claim against the company's future cash flow but do not hold traditional interest-bearing debt or common stock. Nonequity claims include debt equivalents (e.g., operating leases and unfunded pension liabilities) and hybrid securities (e.g., convertible debt and employee options).

EXHIBIT 10.2 Enterprise Valuation of a Single-Business Company

\$ million



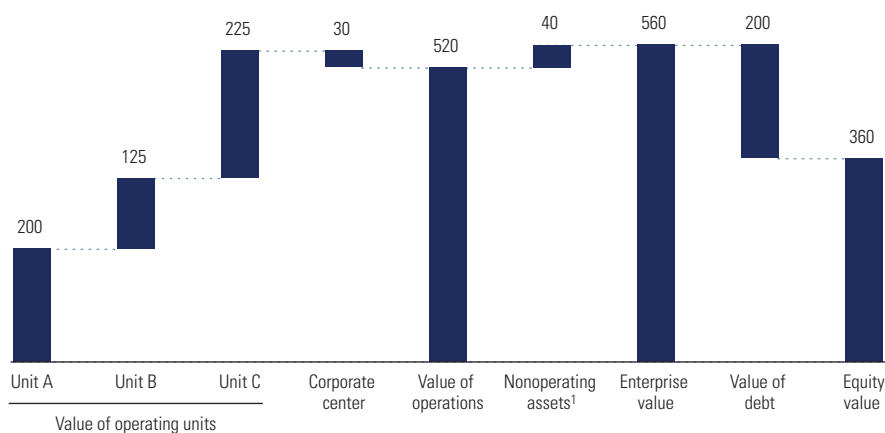
¹ Debt value equals discounted after-tax cash flow to debt holders plus the present value of interest tax shield.

value either directly at \$227.5 million or by estimating enterprise value (\$427.5 million) and subtracting the value of debt (\$200.0 million).

The enterprise DCF method is especially useful when applied to a multibusiness company. As Exhibit 10.3 shows, the enterprise value equals the summed value of the individual operating units less the present value of the corporate-center costs, plus the value of nonoperating assets.³ You can use the enterprise DCF model to value individual projects, business units, and even the entire company with a consistent methodology.

EXHIBIT 10.3 Valuation of a Multibusiness Company

\$ million



¹ Including excess cash and marketable securities.

³ Many investment professionals define enterprise value as interest-bearing debt plus the market value of equity minus cash, whereas we define enterprise value as the value of operations plus nonoperating assets. The investment banker's definition of enterprise value resembles our definition of the value of operations, but only for companies that do not own nonoperating assets (e.g., nonconsolidated subsidiaries) or owe debt equivalents (e.g., unfunded pension liabilities). For companies with significant nonoperating assets or debt equivalents, the banking version of enterprise value can lead to distortions in analysis.

Valuing a company's equity using enterprise DCF is a four-step process:

1. Value the company's operations by discounting free cash flow at the weighted average cost of capital.
2. Identify and value nonoperating assets, such as excess cash and marketable securities, nonconsolidated subsidiaries, and other nonoperating assets not incorporated into free cash flow. Summing the value of operations and nonoperating assets gives enterprise value.⁴
3. Identify and value all debt and other nonequity claims against the enterprise value. Debt and other nonequity claims include fixed-rate and floating-rate debt, debt equivalents such as unfunded pension liabilities and restructuring provisions, employee options, and preferred stock, which are discussed in Chapter 16.
4. Subtract the value of debt and other nonequity claims from enterprise value to determine the value of common equity. To estimate value per share, divide equity value by the number of current shares outstanding.

Exhibit 10.4 presents the results of an enterprise DCF valuation for GlobalCo, an imaginary international logistics company. GlobalCo is used throughout the chapter to compare valuation methods. GlobalCo is a simplified example that ignores the complexities of modern companies. In Appendix H, we present a complete valuation of the global retailer Costco Wholesale.

EXHIBIT 10.4 **GlobalCo: Enterprise DCF Valuation**

\$ million, except where noted

Forecast year	Free cash flow (FCF)	Discount factor at 7.8%	Present value of FCF
Year 1	(2.0)	0.928	(1.9)
Year 2	22.5	0.861	19.4
Year 3	54.6	0.798	43.6
Continuing value	1,176.2	0.798	938.9
Value of operations			1,000.0
Value of nonoperating assets			—
Enterprise value			1,000.0
Less: Value of debt			(250.0)
Less: Debt equivalents and noncontrolling interests			—
Equity value			750.0
Shares outstanding, million			12.5
Equity value per share, \$			60.00

⁴Many investment professionals do not include excess cash when estimating enterprise value and instead net excess cash directly against debt.

We use Costco throughout Part Two to demonstrate in greater detail various parts of the valuation process.

To value GlobalCo, we forecast three years of cash flow. Cash flows generated beyond year 3 are valued using the key value driver formula and reported as continuing value. Next, discount each year's projected free cash flow and the continuing value by the company's weighted average cost of capital.⁵ Sum the present values of the annual cash flows and discounted continuing value to determine the present value of operations.

For simplicity, the first year's projected cash flow is discounted by one full year, the second by two full years, and so on. For the purpose of clear exposition, we assume cash flows occur in lump sums. In actuality, cash flows occur throughout the year, not as a lump sum. Therefore, adjust the discount rate as necessary to better match the timing of cash flows.⁶ The resulting present value is known as the value of operations, which equals \$1 billion for GlobalCo.

To the value of operations, add nonoperating assets, such as excess cash and noncontrolling interests in other companies. Since GlobalCo has no nonoperating assets, the value of operations equals enterprise value. To determine equity value, subtract the value of debt and other nonequity claims. GlobalCo has \$110 million in short-term debt and \$140 million in long-term debt, for a total debt of \$250 million. The company has no unfunded pension obligations or noncontrolling interests held by other companies, but if it did, their value would be subtracted as well.⁷ Divide the resulting equity value of \$750 million by the number of shares outstanding (12.5 million) to estimate a per-share intrinsic value of \$60.

Over the course of the next few sections, we dig deeper into the inputs and the valuation process. Although this chapter presents the enterprise DCF valuation sequentially, valuation is an iterative process.

Valuing Operations

The value of operations equals the discounted value of future free cash flow. Free cash flow equals the cash flow generated by the company's operations, less any reinvestment back into the business. As defined at the beginning of this section, free cash flow is the cash flow available to all investors—equity holders, debt holders, and any other investors—so it is independent of how

⁵ To generate identical results across valuation methods, we have not adjusted figures to eliminate rounding errors. Rounding errors occur in most exhibits.

⁶ If cash flow occurs smoothly throughout the year, lower each discount factor by half a year. If cash flow is heavily weighted toward the year end, as in retail, a smaller adjustment to the discount factor is required. For more on this issue and how to value a company in between fiscal years, see Chapter 16.

⁷ A noncontrolling interest arises when an outside investor owns a minority share of a subsidiary. Since this outside investor has a claim on cash flows, the claim's value must be deducted from enterprise value to compute equity value.

EXHIBIT 10.5 **GlobalCo: Income and Shareholders' Equity Statements**

\$ million

	Historical	Forecast			Continuing value
		Year 1	Year 2	Year 3	
Revenue	200.0	250.0	287.5	301.9	308.5
Operating costs	(120.0)	(150.0)	(172.5)	(181.1)	(185.1)
Depreciation	(20.0)	(25.0)	(28.8)	(30.2)	(30.9)
Operating profit	60.0	75.0	86.3	90.6	92.6
Interest expense	(9.0)	(10.0)	(10.8)	(11.4)	(11.8)
Earnings before taxes	51.0	65.0	75.5	79.1	80.8
Income taxes	(10.2)	(13.0)	(15.1)	(15.8)	(16.2)
Net income	40.8	52.0	60.4	63.3	64.6
Statement of shareholders' equity					
Equity, beginning of year	65.0	98.0	140.0	171.1	
Net income	40.8	52.0	60.4	63.3	
Dividends	(7.8)	(10.0)	(14.3)	(24.1)	
Share issuances (repurchases)	—	—	(15.0)	(30.0)	
Equity, end of year	98.0	140.0	171.1	180.3	

the company is financed. Consistent with this definition, free cash flow must be discounted using the weighted average cost of capital, because the WACC represents rates of return required by the company's debt and equity holders blended together. It is the company's opportunity cost of funds.

Reorganizing the Financial Statements To begin the valuation process, collect the company's historical financial statements. In Exhibit 10.5, we present the income statement and statement of shareholders' equity for GlobalCo. Exhibit 10.6 presents the company's balance sheet. For ease of exposition, we present only one historical year of financial statements. In practice, collect multiple years in order to better assess the long-run performance and future potential of the company.

Although ROIC and free cash flow (FCF) are central to the valuation process, the two measures cannot be computed easily from a company's financial statements, which commingle operating performance and capital structure. Therefore, to calculate ROIC and FCF, first reorganize the accounting financial statements into new statements that clearly separate operating items, nonoperating items, and sources of financing.

This reorganization leads to two new terms: invested capital and net operating profit after taxes (NOPAT). Invested capital represents the investor capital required to fund operations, without distinguishing how the capital is financed. NOPAT represents the total after-tax operating income generated by the company's invested capital, available to all investors. We briefly summarize the reorganization process next, but for a more detailed discussion using Costco, see Chapter 11.

EXHIBIT 10.6 **GlobalCo: Balance Sheet**

\$ million

	Historical	Forecast		
		Year 1	Year 2	Year 3
Cash	4.0	5.0	5.8	6.0
Accounts receivable	20.0	25.0	28.8	30.2
Inventories	40.0	50.0	57.5	60.4
Current assets	64.0	80.0	92.0	96.6
Property and equipment	200.0	250.0	287.5	301.9
Goodwill and acquired intangibles	100.0	100.0	100.0	100.0
Total assets	364.0	430.0	479.5	498.5
Liabilities and equity				
Short-term debt	110.0	110.0	125.4	134.0
Accounts payable	16.0	20.0	23.0	24.2
Current liabilities	126.0	130.0	148.4	158.2
Long-term debt	140.0	160.0	160.0	160.0
Shareholders' equity	98.0	140.0	171.1	180.3
Total liabilities and equity	364.0	430.0	479.5	498.5

In Exhibit 10.7, we reorganize the income statement into NOPAT. To estimate NOPAT, deduct only operating costs and depreciation from revenue. Do not deduct interest expense or add nonoperating income; they will be analyzed and valued separately as part of nonoperating assets and debt, respectively. Operating taxes are computed on operating profit and represent the level of taxes that would be paid if the firm were financed entirely by equity and held only operating assets. A robust valuation will reconcile net income

EXHIBIT 10.7 **GlobalCo: Net Operating Profit after Taxes (NOPAT)**

\$ million

	Historical	Forecast			Continuing value
		Year 1	Year 2	Year 3	
Revenue	200.0	250.0	287.5	301.9	308.5
Operating costs	(120.0)	(150.0)	(172.5)	(181.1)	(185.1)
Depreciation	(20.0)	(25.0)	(28.8)	(30.2)	(30.9)
Operating profit	60.0	75.0	86.3	90.6	92.6
Operating taxes	(12.0)	(15.0)	(17.3)	(18.1)	(18.5)
NOPAT	48.0	60.0	69.0	72.5	74.0
Reconciliation to net income					
Net income	40.8	52.0	60.4	63.3	64.6
Interest expense	9.0	10.0	10.8	11.4	11.8
Interest tax shield	(1.8)	(2.0)	(2.2)	(2.3)	(2.4)
NOPAT	48.0	60.0	69.0	72.5	74.0

EXHIBIT 10.8 **GlobalCo: Invested Capital and Total Funds Invested**

\$ million

	Historical	Forecast		
		Year 1	Year 2	Year 3
Working capital	48.0	60.0	69.0	72.5
Property, plant, and equipment, net	200.0	250.0	287.5	301.9
Invested capital, excluding goodwill	248.0	310.0	356.5	374.3
Goodwill and acquired intangibles	100.0	100.0	100.0	100.0
Invested capital, including goodwill	348.0	410.0	456.5	474.3
Nonoperating assets	—	—	—	—
Total funds invested	348.0	410.0	456.5	474.3
Reconciliation of total funds invested				
Short-term debt	110.0	110.0	125.4	134.0
Long-term debt	140.0	160.0	160.0	160.0
Debt and debt equivalents	250.0	270.0	285.4	294.0
Shareholders' equity	98.0	140.0	171.1	180.3
Total funds invested	348.0	410.0	456.5	474.3

to NOPAT. The reconciliation will prevent unintended errors and force explicit choices about how each piece of data will be incorporated in the valuation.

In Exhibit 10.8, we reorganize the balance sheet into invested capital and total funds invested. Invested capital includes working capital, property, plant, equipment, and other operating assets, net of other operating liabilities. Measure invested capital both including and excluding goodwill and acquired intangibles. By analyzing invested capital with and without goodwill, we can assess the impact of acquisitions on past performance. A company with robust margins and lean operations can have low ROIC with goodwill because of the high prices it paid for acquisitions.

GlobalCo holds only operating assets, so invested capital equals total funds invested. Since nonoperating assets are typically valued using methods other than DCF, we explicitly distinguish them from operating assets and operating liabilities. Next, reconcile total funds invested with sources of capital: debt, equity, and their equivalents. Examples of debt equivalents include unfunded pension obligations and environmental-remediation liabilities. Examples of equity equivalents include deferred taxes.

To calculate ROIC in year 1, divide NOPAT by the prior-year invested capital.⁸ In year 1, ROIC excluding goodwill equals 24.2 percent ($\$60/\248), and ROIC including goodwill equals 17.2 percent ($\$60/\348). Because ROIC

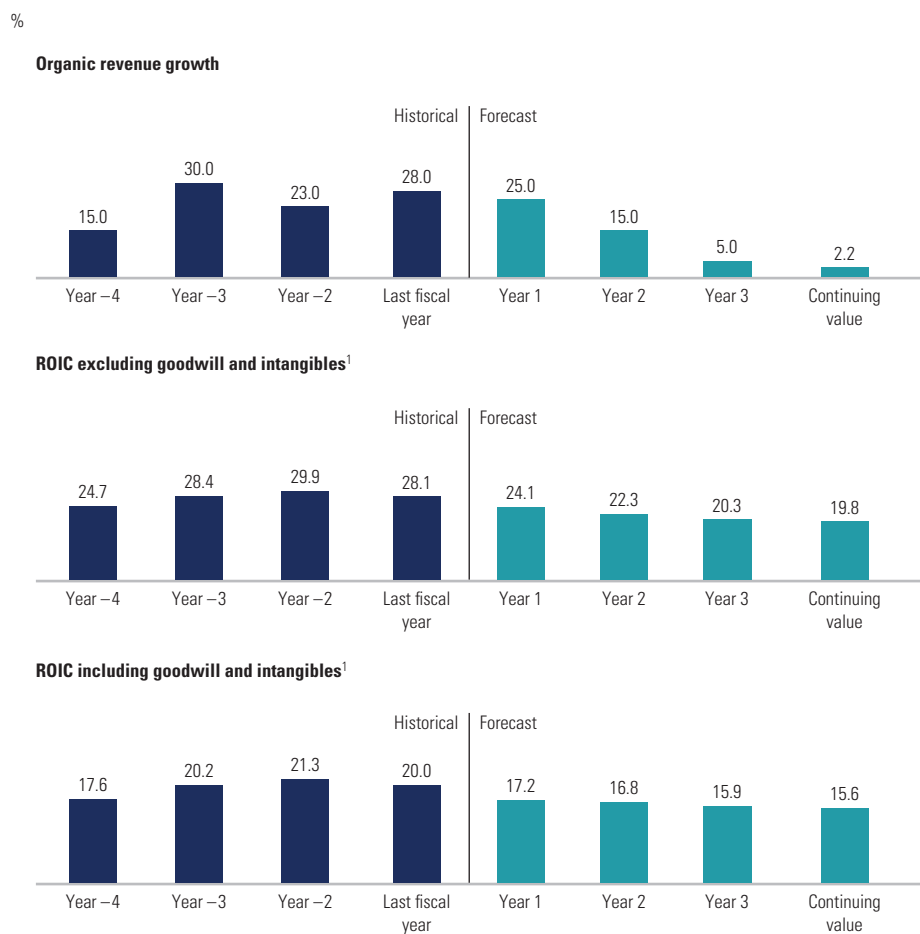
⁸ In this calculation, we estimate ROIC using prior-year invested capital (that is, measured at the beginning of the year) to link our enterprise DCF valuation with an economic profit valuation presented later in this chapter. When benchmarking performance, use a two-year average of invested capital.

is greater than the cost of capital of 7.8 percent, the company is creating value, both with and without the effect of acquisition premiums.

Analyzing Historical Performance Once the company's financial statements are reorganized into NOPAT and invested capital, analyze the company's historical financial performance. By thoroughly analyzing the past, we can understand whether the company has created value, how fast it has grown, and how it compares with its competitors. A good analysis will focus on the key drivers of value: return on invested capital, revenue growth, and free cash flow. Understanding how these drivers behaved in the past will help you make more reliable estimates of future cash flow.

Exhibit 10.9 presents a historical analysis of organic revenue growth and ROIC. GlobalCo has been performing well, with organic growth rates and

EXHIBIT 10.9 **GlobalCo: Forecast Revenue Growth and ROIC**



¹ Measured using beginning-of-year capital to match economic-profit models presented later in the chapter.

ROICs without goodwill both above 20 percent. A good analysis will assess many years—even decades—of past performance. While analysis from long ago may be outdated, understanding how the company performs in different phases of the economic cycle will better inform your forecasts. For an in-depth discussion of financial analysis using reorganized financial statements, see Chapter 12.

Projecting Revenue Growth, ROIC, and Free Cash Flow Based on insights from your historical analysis, as well as forecasts of economic and industry trends, create a set of integrated financial statements going forward. In Exhibits 10.5 and 10.6, we present line-by-line forecasts of the income statement, statement of shareholders' equity, and balance sheet. The three statements should be integrated in that net income should flow through the statement of equity, which should match the corresponding account in the balance sheet. Use excess cash, debt, dividends, or a combination thereof to ensure that the balance sheet balances. Chapter 13 provides details on the forecasting process.

When building the forecast model, use judgment on how much detail to forecast at various points. Over the short run (the first few years), forecast each financial-statement line item, such as gross margin, selling expenses, accounts receivable, and inventory. This will allow you to incorporate visible trends in individual line items. Moving further out, individual line items become difficult to project, and a high level of detail can obscure the critical value drivers. Therefore, over the medium horizon (5 to 15 years), focus on the company's key value drivers, such as operating margin, the operating tax rate, and capital efficiency. At some point, projecting even key drivers on a year-by-year basis becomes impractical. To value cash flows beyond this point, use a continuing-value formula, often called the terminal value. Choosing an appropriate point of transition depends on the company and how it is changing over time. A company undergoing significant change may require a long, detailed window, whereas a stable, mature company may require very little detail in your forecasts.

Next, use the reorganized financial statements to calculate free cash flow. Exhibit 10.10 presents the free cash flow for GlobalCo. Defined in a manner consistent with ROIC, free cash flow is derived directly from NOPAT and the change in invested capital. Unlike the accounting statement of cash flows (provided in the company's annual report), free cash flow is independent of nonoperating items and capital structure.

Estimating Continuing Value At the point where predicting the individual key value drivers on a year-by-year basis becomes impractical, do not vary the individual drivers over time. Instead, use a perpetuity-based continuing value, such that:

EXHIBIT 10.10 **GlobalCo: Projected Free Cash Flow**

\$ million

	Year 1	Year 2	Year 3
NOPAT	60.0	69.0	72.5
Depreciation	20.0	25.0	28.8
Gross cash flow	80.0	94.0	101.2
Decrease (increase) in operating working capital	(12.0)	(9.0)	(3.4)
Capital expenditures, net of disposals	(70.0)	(62.5)	(43.1)
Free cash flow	(2.0)	22.5	54.6
Reconciliation of free cash flow			
Interest expense	10.0	10.8	11.4
Interest tax shield	(2.0)	(2.2)	(2.3)
Decrease (Increase) in short-term debt	0.0	(15.4)	(8.6)
Decrease (Increase) in long-term debt	(20.0)	—	—
Flows to (from) debt holders	(12.0)	(6.8)	0.5
Cash dividends	10.0	14.3	24.1
Repurchased (issued) shares	—	15.0	30.0
Flows to (from) equity holders	10.0	29.3	54.1
Free cash flow	(2.0)	22.5	54.6

$$\begin{aligned} & \text{Present Value of Free Cash Flow} \\ & \text{during Explicit Forecast Period} \\ \text{Value of Operations} = & \quad + \\ & \text{Present Value of Free Cash Flow} \\ & \text{after Explicit Forecast Period} \end{aligned}$$

Although many continuing-value models exist, we prefer the key value driver formula presented in Chapter 3. The key value driver formula is superior to alternative methodologies because it is based on cash flow and it links cash flow directly to growth and ROIC. The key value driver formula is expressed as follows:

$$\text{Continuing Value}_t = \frac{\text{NOPAT}_{t+1} \left(1 - \frac{g}{\text{RONIC}} \right)}{\text{WACC} - g}$$

The formula requires a forecast of NOPAT in the year following the explicit forecast period, the long-run forecast for return on new invested capital (RONIC) purchased during the continuing value period, the WACC, and long-run growth (g) in NOPAT.

Exhibit 10.11 presents an estimate for GlobalCo's continuing value. Based on a final-year estimate of NOPAT from Exhibit 10.7 of \$74.0 million, RONIC excluding goodwill from Exhibit 10.9 of 19.8 percent, and a long-term growth

EXHIBIT 10.11 **GlobalCo: Continuing Value**

\$ million

Key inputs¹

Projected NOPAT in final forecast year	74.0
NOPAT growth rate in perpetuity (<i>g</i>)	2.2%
Return on new invested capital (RONIC)	19.8%
Weighted average cost of capital (WACC)	7.8%

$$\begin{aligned}\text{Continuing value}_t &= \frac{\text{NOPAT}_{t+1} \left(1 - \frac{g}{\text{RONIC}} \right)}{\text{WACC} - g} \\ &= \$1,176.2^1\end{aligned}$$

¹ \$1,176.2 is calculated from unrounded data. Rounded inputs calculate to \$1,174.6 million.

rate from Exhibit 10.9 of 2.2 percent, the continuing value is estimated at \$1,176.2 million. The valuation model presented in Exhibit 10.4 discounts this value into today’s dollars and adds it to the value from the explicit forecast period to determine GlobalCo’s operating value.

Alternative methods and additional details for estimating continuing value are provided in Chapter 14.

Discounting Free Cash Flow at the Weighted Average Cost of Capital In an enterprise valuation, free cash flows are available to all investors. Consequently, the discount factor for free cash flow must represent the risk faced by all investors. The WACC blends the rates of return required by debt holders (k_d) and equity holders (k_e). For a company financed solely with debt and equity, the WACC is defined as follows:

$$\text{WACC} = \frac{D}{D + E} k_d (1 - T_m) + \frac{E}{D + E} k_e$$

where debt (D) and equity (E) are measured using market values. Note how the cost of debt has been reduced by the marginal tax rate (T_m). The reason for doing this is that the tax shield attributable to interest has been excluded from free cash flow. Since the interest tax shield (ITS) has value to the shareholder, it must be incorporated in the valuation. Enterprise DCF values the tax shield by reducing the weighted average cost of capital.

Why move interest tax shields from free cash flow to the cost of capital? By calculating free cash flow as if the company were financed entirely with equity, one can compare operating performance across companies and over time without regard to capital structure. By focusing solely on operations, it is possible to develop a clearer picture of historical performance, and this leads to better performance measurement and forecasting.

Although applying the WACC is intuitive and relatively straightforward, it has some drawbacks. If you discount all future cash flows with a constant cost of capital, as most analysts do, you are implicitly assuming the company keeps its capital structure constant at a target ratio of debt to equity. But if a company plans, say, to increase (or decrease) its debt-to-value ratio, the

EXHIBIT 10.12 **GlobalCo: Weighted Average Cost of Capital**

%					
Source of capital	Proportion of total capital	Cost of capital	Marginal tax rate	After-tax cost of capital	Contribution to weighted average
Debt	25.0	4.0	20.0	3.2	0.8
Equity	75.0	9.3		9.3	7.0
WACC	100.0				7.8

current cost of capital will understate (or overstate) the expected tax shields. The WACC can be adjusted to accommodate a changing capital structure. However, the process is complicated, and in these situations, we recommend an alternative method such as adjusted present value (APV).

The weighted average cost of capital for GlobalCo is presented in Exhibit 10.12. GlobalCo's 7.8 percent WACC is based on a cost of equity of 9.3 percent, pretax cost of debt of 4.0 percent, and a debt-to-value ratio of 25 percent.

Identifying and Valuing Nonoperating Assets

Many companies own assets that have value but whose cash flows are not included in accounting revenue or operating profit. As a result, the cash generated by these assets is not part of free cash flow and must be valued separately.

For example, consider equity investments, known outside the United States as nonconsolidated subsidiaries. When a company owns a minority stake in another company, it will not record the company's revenue or costs as part of its own. Instead, the company will record only its proportion of the other company's net income as a separate line item.⁹ Including net income from nonconsolidated subsidiaries as part of the parent's operating profit will distort margins, since only the subsidiaries' profit is recognized and not the corresponding revenues. Consequently, nonconsolidated subsidiaries are best analyzed and valued separately.

Other nonoperating assets include excess cash, tradable securities, and customer-financing business units. A detailed process for identifying and valuing nonoperating assets appears in Chapter 16.

Identifying and Valuing Debt and Other Nonequity Claims

To convert enterprise value into equity value, subtract debt and other non-equity claims, such as unfunded retirement liabilities, capitalized operat-

⁹ For stakes between 20 percent and 50 percent, the parent company will recognize its proportion of the subsidiary's income. A parent that owns less than a 20 percent stake in another company records only dividends paid as part of its own income. This makes valuation of stakes of less than 20 percent in privately held companies extremely challenging.

ing leases, and outstanding employee options. Common equity is a residual claimant, receiving cash flows only *after* the company has fulfilled its other contractual claims. Careful analysis of all potential claims against cash flows is therefore critical.

Nonequity claims on a company's cash flow are not always easy to spot. Many of the accounting scandals that led to the Sarbanes-Oxley legislation in the United States involved undisclosed or carefully hidden liabilities. Even more than a decade later, hidden liabilities remain an issue for investors. For instance, Netflix was accused in 2012 of failing to disclose \$3.7 billion in contractual promises to production companies.¹⁰ In this case, these promises were operating related and would already be incorporated into projections of free cash flow. Nonetheless, these promises are a priority claim on the company's assets and need to be assessed accordingly.

Although a comprehensive list of nonequity claims is impractical, here are the most common:

- *Debt.* If available, use the market value of all outstanding debt, including fixed- and floating-rate debt. If that information is unavailable, the book value of debt is a reasonable proxy, unless the probability of default is high or interest rates have changed dramatically since the debt was originally issued. Any valuation of debt, however, should be consistent with your estimates of enterprise value. (See Chapter 16 for more details.)
- *Leases.* Rather than purchase assets outright, many companies lease certain assets for a fixed period. Any lease payments recorded as interest expense and not rental expense must be valued separately and deducted from enterprise value.
- *Unfunded retirement liabilities.* Companies with defined-benefit pension plans and promised retiree medical benefits may have unfunded obligations that should be treated like debt.
- *Preferred stock.* For large stable companies, preferred stock more closely resembles unsecured debt. For small start-ups, preferred stock contains valuable options. In both situations, value preferred stocks separately from common stock.
- *Employee options.* Many companies offer their employees compensation in the form of options. Since options give the employee the right to buy company stock at a discounted price, they can have great value and must also be factored into equity value.
- *Noncontrolling interests.* When a company has majority control of a subsidiary but does not own 100 percent, the entire subsidiary must be consolidated on the parent company's balance sheet. The funding other investors provide for this subsidiary is recognized on the parent

¹⁰ Cory Johnson, "The Scary \$3B Bomb Not on Netflix's Balance Sheet," Bloomberg TV, July 5, 2012, www.bloomberg.com.

company's balance sheet as noncontrolling interests (formerly called minority interest). When valuing noncontrolling interests, it is important to realize that the minority interest holder does not have a claim on the company's assets, but rather a claim on the subsidiary's assets.

The identification and valuation of nonequity claims are covered in detail in Chapter 16. A detailed discussion of how to analyze leases is presented in Chapter 22. Additional detail on retirement obligations can be found in Chapter 23.

A common mistake made when valuing companies is to double-count nonequity claims already deducted from cash flow. Consider a company with a pension shortfall. You have been told the company will make extra payments to eliminate the liability. If you deduct the present value of the liability from enterprise value, do not model the extra payments within free cash flow; that would mean double-counting the shortfall (once in cash flow and once as a debtlike claim), leading to an underestimate of equity value.

Valuing Equity

Once you have identified and valued all nonequity claims, subtract the value of these claims from enterprise value to determine equity value. In Exhibit 10.4, we subtract \$250 million in debt, both short-term and long-term debt, from \$1 billion in enterprise value. Since GlobalCo has no debt equivalents, this leads to an intrinsic equity value of \$750 million.

To determine GlobalCo's share price, divide the intrinsic equity value by the number of *undiluted* shares outstanding. Do not use diluted shares. Convertible debt, convertible preferred stock, and employee stock options should be valued separately. If you were to subtract the value of these claims and use diluted shares, you would double-count the options' value. At the time of GlobalCo's valuation, the company had 12.5 million shares outstanding. Dividing the equity estimate of \$750 million by 12.5 million shares generates an estimated value of \$60 per share.

Although it appears the valuation is complete, the job is not done. Compare the intrinsic value with market prices. If the two values differ, and they probably will, search for the cause, such as overly optimistic forecasts or missing liabilities. Next, use the model to test the sensitivity of various inputs on the valuation. Determine which inputs lead to the biggest changes, and which lead to negligible differences. Use this analysis to identify opportunities, prioritize operating activities, and identify risks.

ECONOMIC PROFIT-BASED VALUATION MODELS

The enterprise DCF model is a favorite of academics and practitioners because it relies solely on how cash flows in and out of the company. Complex accounting can be replaced with a simple question: Does cash change hands? One shortfall of enterprise DCF, however, is that each year's cash flow provides

little insight into the company's competitive position and economic performance. Declining free cash flow can signal either poor performance or investment for the future. The economic-profit model highlights how and when the company creates value, yet properly implemented, it leads to a valuation that is identical to that of enterprise DCF.

Economic profit measures the value created by the company in a single period and is defined as follows:

$$\text{Economic Profit} = \text{Invested Capital} \times (\text{ROIC} - \text{WACC})$$

Since ROIC equals NOPAT divided by invested capital, we can rewrite the equation as follows:

$$\text{Economic Profit} = \text{NOPAT} - (\text{Invested Capital} \times \text{WACC})$$

Exhibit 10.13 presents economic-profit calculations for GlobalCo using both methods. Not surprisingly, with an ROIC more than double its cost of capital, GlobalCo generates significant economic profits.

To demonstrate how economic profit can be used to value a company—and to demonstrate its equivalence to enterprise DCF—consider a stream of growing cash flows valued using the growing-perpetuity formula:

$$\text{Value}_0 = \frac{\text{FCF}_1}{\text{WACC} - g}$$

In Chapter 3, we transformed this cash flow perpetuity into the key value driver model. The key value driver model is superior to the simple cash flow perpetuity model, because it explicitly models the relationship between growth and required investment. Using a few additional algebraic steps (detailed in Appendix A) and the assumption that the company's ROIC on new projects equals the ROIC on existing capital, it is possible to transform the cash flow perpetuity into a key value driver model based on economic profits:

$$\text{Value}_0 = \text{Invested Capital}_0 + \frac{\text{Invested Capital}_0 \times (\text{ROIC}_1 - \text{WACC})}{\text{WACC} - g}$$

Finally, we substitute the definition of economic profit:

$$\text{Value}_0 = \text{Invested Capital}_0 + \frac{\text{Economic Profit}_1}{\text{WACC} - g}$$

As can be seen in the economic-profit-based key value driver model, the operating value of a company equals its book value of invested capital plus the present value of all future value created. In this case, the future economic

EXHIBIT 10.13 GlobalCo: Economic-Profit Summary

\$ million, except where noted

	Year 1	Year 2	Year 3
Method 1			
Return on invested capital, ¹ %	24.2	22.3	20.3
Weighted average cost of capital, %	(7.8)	(7.8)	(7.8)
Economic spread, %	16.4	14.5	12.5
× Invested capital ¹	248.0	310.0	356.5
= Economic profit	40.7	44.8	44.6
Method 2			
Invested capital ¹	248	310	357
× Weighted average cost of capital, %	7.8	7.8	7.8
= Capital charge	19.3	24.2	27.8
NOPAT	60.0	69.0	72.5
Capital charge	(19.3)	(24.2)	(27.8)
Economic profit	40.7	44.8	44.6
Including goodwill			
Economic profit including goodwill	32.9	37.0	36.8

¹ Invested capital measured at the beginning of the year, excluding goodwill and acquired intangibles.

profits are valued using a growing perpetuity, because the company's economic profits are increasing at a constant rate over time. The formula also demonstrates that when economic profit is expected to be zero, the value of operations will equal invested capital. If a company's value of operations exceeds its invested capital, be sure to identify the sources of competitive advantage that allows the company to maintain superior financial performance.

More generally, economic profit can be valued as follows:

$$\text{Value}_0 = \text{Invested Capital}_0 + \sum_{t=1}^{\infty} \frac{\text{Economic Profit}_t}{(1 + \text{WACC})^t}$$

Since the economic-profit valuation was derived directly from the free cash flow model (see Appendix A for a general proof of equivalence), any valuation based on discounted economic profits will be identical to enterprise DCF. To assure equivalence, however, it is necessary to do the following:

- Use beginning-of-year invested capital (i.e., last year's value) instead of average or current-year invested capital, which is common to competitive benchmarking.
- Define invested capital for both economic profit and ROIC using the same value. For example, ROIC can be measured either with or without

goodwill. If you measure ROIC without goodwill, you must also measure invested capital without goodwill. All told, it doesn't matter how you define invested capital, as long as you are consistent.

- Use a constant cost of capital to discount projections.

Exhibit 10.14 presents the valuation results for GlobalCo using discounted economic profit. Economic profits are explicitly forecast for three years; the remaining years are valued using an economic-profit continuing-value formula.¹¹ Comparing the equity value from Exhibit 10.4 with that of Exhibit 10.14, we see that the estimate of GlobalCo's DCF value is the same, regardless of the method.

The benefits of economic profit become apparent when we examine the drivers of economic profit, ROIC and WACC, on a year-by-year basis in Exhibit 10.14. Note that the valuation is contingent on returns that exceed the

EXHIBIT 10.14 **GlobalCo: Valuation Using Discounted Economic Profit**

\$ million, except where noted

Year	Invested capital ¹	ROIC, ¹ %	WACC, %	Economic profit	Discount factor at 7.8%	Present value of economic profit
Year 1	348.0	17.2	(7.8)	32.9	0.928	30.5
Year 2	410.0	16.8	(7.8)	37.0	0.861	31.9
Year 3	456.5	15.9	(7.8)	36.8	0.798	29.4
Continuing value				701.8	0.798	560.3
Present value of economic profit						652.0
Invested capital including goodwill ¹						348.0
Value of operations						1,000.0
Nonoperating assets						—
Enterprise value						1,000.0
Less: Value of debt						(250.0)
Less: Value of noncontrolling interest						—
Equity value						750.0

¹ Invested capital measured at the beginning of the year with goodwill and acquired intangibles.

¹¹ To calculate continuing value, you can use the economic-profit-based key value driver formula, but only if RONIC equals ROIC in the continuing-value year. If RONIC going forward differs from the final year's ROIC, then the equation must be separated into current and future economic profits:

$$\text{Value}_t = \text{IC}_t + \frac{\text{IC}_t (\text{ROIC}_{t+1} - \text{WACC})}{\text{WACC}} + \frac{\text{PV}(\text{Economic Profit}_{t+2})}{\text{WACC} - g}$$

Current Economic Profits Future Economic Profits

such that:

$$\text{PV}(\text{Economic Profit}_{t+2}) = \frac{\text{NOPAT}_{t+1} \left(\frac{g}{\text{RONIC}} \right) (\text{RONIC} - \text{WACC})}{\text{WACC}}$$

For more on these and other continuing value formulas, see Chapter 14.

cost of capital but drop over time as new competitors enter and put pressure on operating margins. Explicitly modeling ROIC as a primary driver of economic profit prominently displays expectations of value creation. Conversely, the FCF model fails to highlight when a company creates and destroys value. Free cash flow combines ROIC and growth, two critical but very different value drivers.

Also note how GlobalCo's high ROIC—double its cost of capital—leads to an operating value that exceeds the book value of its invested capital (\$1 billion versus \$348 million). When investors believe a company will create value, enterprise value will be greater than invested capital.

ADJUSTED-PRESENT-VALUE MODEL

When building an enterprise DCF or economic-profit valuation, most investment professionals discount all future flows at a constant WACC. Using a constant WACC, however, assumes the company manages its capital structure to a target debt-to-value ratio.

In most situations, debt grows with company value. But suppose the company planned to change its capital structure significantly, as in a leveraged buyout. Indeed, companies with a high proportion of debt often pay it down as cash flow improves, thus lowering their future debt-to-value ratios. In these cases, a valuation based on a constant WACC would overstate the value of the tax shields. Although the WACC can be adjusted yearly to handle a changing capital structure, the process is complex. Therefore, we turn to the most flexible of valuation models: adjusted present value (APV).

The APV model separates the value of operations into two components: the value of operations as if the company were all-equity financed and the value of tax shields that arise from debt financing:¹²

$$\text{Adjusted Present Value} = \text{Enterprise Value as if the Company Were All-Equity Financed} + \text{Present Value of Tax Shields}$$

The APV valuation model follows directly from the teachings of economists Franco Modigliani and Merton Miller, who proposed that in a market with no taxes (among other things), a company's choice of financial structure will not affect the value of its economic assets. Only market imperfections, such as taxes and distress costs, affect enterprise value.

When building a valuation model, it is easy to forget these teachings. To see this, imagine a company (in a world with no taxes) that has a 50/50 mix

¹² This book focuses on the tax shields generated by interest expense. On a more general basis, the APV values any cash flows associated with capital structure, such as tax shields, issuance costs, and distress costs. Distress costs include direct costs, such as court-related fees, and indirect costs, such as the loss of wary customers and suppliers.

of debt and equity. If the company's debt has an expected return of 5 percent and the company's equity has an expected return of 15 percent, its weighted average cost of capital would be 10 percent. Suppose the company decides to issue more debt, using the proceeds to repurchase shares. Since the cost of debt is lower than the cost of equity, it would appear that issuing debt to retire equity should lower the WACC, raising the company's value.

This line of thinking is flawed, however. In a world without taxes, a change in capital structure would not change the cash flow generated by operations, nor the risk of those cash flows. Therefore, neither the company's enterprise value nor its cost of capital would change. So why would we think it would? When adding debt, we adjusted the weights, but we failed to properly increase the cost of equity. Since debt payments have priority over cash flows to equity, adding leverage increases the risk to equity holders. When leverage rises, they demand a higher return. Modigliani and Miller postulated that this increase would perfectly offset the change in weights.

In reality, taxes play a role in determining capital structure. Since interest is tax deductible, profitable companies can lower taxes by raising debt. But if the company relies too heavily on debt, the company's customers and suppliers may fear financial distress and be reluctant to do business with the company, reducing future cash flow (academics call this distress costs or dead-weight costs). Rather than model the effect of capital-structure changes in the weighted average cost of capital, APV explicitly measures and values the cash flow effects of financing separately.

To build an APV valuation, value the company as if it were all-equity financed. Do this by discounting free cash flow by the unlevered cost of equity (what the cost of equity would be if the company had no debt).¹³ To this value, add any value created by the company's use of debt. Exhibit 10.15 values GlobalCo using adjusted present value.

Since we assume (for expositional purposes) that GlobalCo will manage its capital structure to a target debt-to-value level of 25 percent, the APV-based valuation leads to the same value for equity as did enterprise DCF (see Exhibit 10.4) and economic profit (see Exhibit 10.14). A simplified proof of equivalence between enterprise DCF and adjusted present value can be found in Appendix B. The following subsections explain adjusted present value in detail.

Valuing Free Cash Flow at Unlevered Cost of Equity

When valuing a company using the APV, explicitly separate the unlevered value of operations (V_u) from any value created by financing, such as tax

¹³ Free cash flow projections in the APV model are identical to those presented in Exhibit 10.4. Continuing value is computed using the key value driver formula. Only the cost of capital is used for discounting changes.

EXHIBIT 10.15 GlobalCo: Valuation Using Adjusted Present Value

\$ million, except where noted

Year	Free cash flow (FCF)	Interest tax shield (ITS)	Discount factor at 8.0%	Present value of FCF	Present value of ITS
Year 1	(2.0)	2.0	0.926	(1.9)	1.9
Year 2	22.5	2.2	0.857	19.3	1.9
Year 3	54.6	2.3	0.794	43.4	1.8
Continuing value	1,135.6	40.6	0.794	901.5	32.2
Present value				962.3	37.7
Present value of free cash flow					962.3
Present value of interest tax shield					37.7
Value of operations					1,000.0

shields (V_{txa}). For a company with debt (D) and equity (E), this relationship is as follows:

$$V_u + V_{txa} = D + E \quad (10.1)$$

A second result of Modigliani and Miller's work is that the total risk of the company's assets, real and financial, must equal the total risk of the financial claims against those assets. Thus, in equilibrium, the blended cost of capital for operating assets (k_u), which we call the unlevered cost of equity) and financial assets (k_{txa}) must equal the blended cost of capital for debt (k_d) and equity (k_e):

$$\frac{V_u}{V} k_u + \frac{V_{txa}}{V} k_{txa} = \frac{D}{V} k_d + \frac{E}{V} k_e \quad (10.2)$$

In the corporate-finance literature, academics combine Modigliani and Miller's two equations to solve for the cost of equity (k_e) in order to demonstrate the relationship between leverage and the cost of equity. Appendix C algebraically rearranges Equation 10.2 to solve for the most flexible version of the levered cost of equity:

$$k_e = k_u + \frac{D}{E} (k_u - k_d) - \frac{V_{txa}}{E} (k_u - k_{txa}) \quad (10.3)$$

As this equation indicates, the cost of equity depends on the unlevered cost of equity, or the cost of equity when the company has no debt, plus a premium for leverage, less a reduction for the tax deductibility of debt. Note that when a company has no debt ($D = 0$) and subsequently no tax shields ($V_{txa} = 0$), k_e equals k_u . This is why k_u is referred to as the unlevered cost of equity.

Unfortunately, k_u cannot be observed directly. In fact, none of the variables on the left side of Equation 10.2 can be observed directly. Only the values on the right—that is, those related to debt and equity—can be estimated using

market data. Because there are so many unknowns and only one equation, we must impose additional restrictions to build a usable relationship between the levered (k_e) and unlevered (k_u) cost of equity.

If you believe the company will manage its debt-to-value ratio to a target level (the company's debt will grow with the business), then the value of the tax shields will track the value of the operating assets. Thus, the risk of tax shields will mirror the risk of operating assets ($k_{txa} = k_u$). Setting k_{txa} equal to k_u , Equation 10.3 can be simplified as follows:

$$k_e = k_u + \frac{D}{E}(k_u - k_d) \quad (10.4)$$

The unlevered cost of equity can now be reverse engineered using the observed cost of equity, the cost of debt, and the market debt-to-equity ratio. (Appendix C shows some alternative versions for deriving k_u from k_e .)

Valuing Tax Shields and Other Capital Structure Effects

To complete an APV valuation, forecast and discount capital structure side effects such as tax shields, security issuance costs, and distress costs. Since GlobalCo has only a small probability of default, we estimated the company's future interest tax shields using the company's expected interest payments and marginal tax rate (see Exhibit 10.16). To calculate the expected interest payment in year 1, multiply the prior year's debt of \$250 million by the interest rate of 4.0 percent. This results in an expected interest payment of \$10 million. Next, multiply the expected interest payment by the marginal tax rate of 20 percent, for an expected interest tax shield of \$2 million in year 1. To determine the continuing value of interest tax shields beyond year 3, use a growth perpetuity based on interest tax shields in the continuing-value year, the unlevered cost of capital, and growth in NOPAT.

A company with significant leverage may not be able to fully use the tax shields (it may not have enough profits to shield). If there is a significant

EXHIBIT 10.16 **GlobalCo: Forecast of Interest Tax Shields**

\$ million

Forecast year	Prior-year net debt ¹	Interest rate, %	Expected interest payment	Marginal tax rate, %	Interest tax shield
Year 1	250.0	4.0	10.0	20.0	2.0
Year 2	270.0	4.0	10.8	20.0	2.2
Year 3	285.4	4.0	11.4	20.0	2.3
Continuing-value forecast	294.0	4.0	11.8	20.0	2.4

¹ Total debt net of excess cash.

probability of default, you must model *expected* tax shields, rather than the calculated tax shields based on promised interest payments.¹⁴ To do this, reduce each promised tax shield by the cumulative probability of default.

CAPITAL CASH FLOW MODEL

When a company actively manages its capital structure to a target debt-to-value level, both free cash flow (FCF) and the interest tax shield (ITS) should be discounted at the unlevered cost of equity, k_u , such that enterprise value equals the sum of discounted cash flows plus the sum of discounted interest tax shields:

$$V = \sum_{t=1}^{\infty} \frac{FCF_t}{(1+k_u)^t} + \sum_{t=1}^{\infty} \frac{ITS_t}{(1+k_u)^t}$$

In 2002, Richard Ruback of the Harvard Business School argued that there is no need to separate free cash flow from tax shields when both flows are discounted by the same cost of capital.¹⁵ He combined the two flows and named the resulting cash flow (i.e., FCF plus interest tax shields) capital cash flow (CCF):

$$V = PV(\text{Capital Cash Flow}) = \sum_{t=1}^{\infty} \frac{FCF_t + ITS_t}{(1+k_u)^t}$$

Given that Ruback's assumptions match those of the weighted average cost of capital, the capital cash flow and WACC-based valuations will lead to identical results. In fact, we have now detailed three distinct but identical valuation methods created solely around how they treat tax shields: WACC (tax shield valued in the cost of capital), APV (tax shield valued separately), and CCF (tax shield valued in the cash flow).

Although free cash flow and capital cash flow lead to the same result when debt is proportional to value, we believe FCF models are superior to CCF models. By keeping NOPAT and FCF independent of leverage, it is easier to evaluate the company's operating performance over time and against competitors. A clean measure of historical operating performance leads to better forecasts.

¹⁴ The Tax Cuts and Jobs Act of 2017 placed additional restrictions on the deductibility of interest, even for profitable companies. Only value interest tax shields if they meet deductibility guidelines.

¹⁵ R. S. Ruback, "Capital Cash Flows: A Simple Approach to Valuing Risky Cash Flows," *Financial Management* (Summer 2002): 85–103.

CASH-FLOW-TO-EQUITY VALUATION MODEL

Each of the preceding valuation models determined the value of equity indirectly by subtracting debt and other nonequity claims from enterprise value. The equity cash flow model values equity directly by discounting cash flows to equity (CFE) at the cost of equity, rather than at the weighted average cost of capital.¹⁶

Exhibit 10.17 details the cash flow to equity for GlobalCo. Cash flow to equity starts with net income. To this, add back noncash expenses to determine gross cash flow. Next, subtract investments in working capital, fixed assets, and nonoperating assets. Finally, add any increases in debt and other nonequity claims, and subtract decreases in debt and other nonequity claims. Unlike free cash flow, cash flow to equity includes operating, nonoperating, and financing items in the calculation. Alternatively, you can compute cash flow to equity as dividends plus share repurchases minus new equity issues. The two methods generate identical results.¹⁷

To value GlobalCo using cash flow to equity holders, discount projected equity cash flows at the cost of equity (see Exhibit 10.18). Unlike enterprise-based models, this method makes no adjustments to the DCF value for nonoperating assets or debt. Rather, they are embedded as part of the equity cash flow.

EXHIBIT 10.17 GlobalCo: Equity Cash Flow Summary

\$ million

	Forecast		
	Year 1	Year 2	Year 3
Net income	52.0	60.4	63.3
Depreciation	20.0	25.0	28.8
Gross cash flow	72.0	85.4	92.1
Decrease (increase) in operating working capital	(12.0)	(9.0)	(3.4)
Capital expenditures, net of disposals	(70.0)	(62.5)	(43.1)
Increase (decrease) in short-term debt	—	15.4	8.6
Increase (decrease) in long-term debt	20.0	—	—
Cash flow to equity holders	10.0	29.3	54.1
Reconciliation of cash flow to equity			
Cash dividends	10.0	14.3	24.1
Repurchased (issued) shares	—	15.0	30.0
Cash flow to equity holders	10.0	29.3	54.1

¹⁶ The equity method can be difficult to implement correctly, because capital structure is embedded in the cash flow, so forecasting is difficult. For companies whose operations are related to financing, such as financial institutions, the equity method is appropriate. Chapter 38 discusses valuing financial institutions.

¹⁷ Calculate the continuing value using an equity-based variant of the key value driver formula:

$$V_e = \frac{\text{Net Income} \left(1 - \frac{g}{\text{ROE}} \right)}{k_e - g}$$

EXHIBIT 10.18 **GlobalCo: Valuation Using Cash Flow to Equity**

\$ million, except where noted

Forecast year	Cash flow to equity (CFE)	Discount factor at 8.9%	Present value of CFE
2014	10.0	0.915	9.1
2015	29.3	0.837	24.5
2016	54.1	0.765	41.4
Continuing value	882.1	0.765	675.0
Present value of equity cash flows			750.0
Less: Value of noncontrolling interest			—
Equity value			750.0

Once again, note how the valuation, derived using equity cash flows, matches each of the prior valuations. This occurs because we have carefully modeled GlobalCo's debt-to-value ratio at a constant level. If leverage is expected to change, the cost of equity must be appropriately adjusted to reflect the change in risk imposed on equity holders. Although formulas exist to adjust the cost of equity (as done in the APV section earlier in this chapter), many of the best-known formulas are built under restrictions that may be inconsistent with the way you are implicitly forecasting the company's capital structure via the cash flows. This will cause a mismatch between cash flows and the cost of equity, resulting in an incorrect valuation.

It is quite easy to change the company's capital structure without realizing it when using the cash-flow-to-equity model—and that is what makes implementing the equity model so risky. Suppose you plan to value a company whose debt-to-value ratio is 25 percent. You believe the company will pay extra dividends, so you increase debt to raise the dividend payout ratio. Presto! Increased dividends lead to higher equity cash flows and a higher valuation. Even though operating performance has not changed, the equity value has mistakenly increased. What is happening? Using new debt to pay dividends causes a rise in the debt-to-value ratio. Unless you adjust the cost of equity, the valuation will rise incorrectly.

A second major shortcoming of the equity cash flow model is how it values nonoperating assets. Imagine a company that holds a significant amount of low-risk, low-return excess cash. Since operating and nonoperating cash flows are combined in cash flows to equity, they will both be discounted at the same rate, the cost of equity. Since the cost of equity exceeds the rate of return on cash, it appears as if the nonoperating asset is destroying value, and the asset will be incorrectly valued below its book value, even if the asset in actuality is earning a fair rate of return.

A third shortcoming of the cash-flow-to-equity model emerges when valuing a company by business unit. The direct equity approach requires allocating debt and interest expense to each unit. This creates extra work yet provides few additional insights.

One situation where the equity cash flow model leads to the simplest implementation is the analysis and valuation of financial institutions. Since capital structure is a critical part of operations in a financial institution, using enterprise DCF to separate operations and capital structure requires unnecessary assumptions. This is why Chapter 38 uses the cash-flow-to-equity model to value banks and other financial institutions.

PROBLEMATIC MODIFICATIONS TO DISCOUNTED CASH FLOW

In this chapter, we valued GlobalCo by discounting nominal cash flows at a nominal cost of capital. An alternative is to value companies by projecting cash flow in real terms, ignoring the rise in prices, and discounting this cash flow at a real discount rate (the nominal rate less expected inflation). But most managers think in terms of nominal rather than real measures, so nominal measures are often easier to communicate. In addition, interest rates are generally quoted nominally rather than in real terms, excluding expected inflation.

A second difficulty occurs when calculating and interpreting ROIC. The historical statements are nominal, so historical returns on invested capital are nominal. But if the projections for the company use real rather than nominal forecasts, returns on new capital are also real. Projected returns on total capital—new and old—are a combination of nominal and real, so they are impossible to interpret. The only way around this is to restate historical performance on a real basis, which is a complex and time-consuming task. The extra insights gained rarely equal the effort, except in extremely high-inflation environments, described in Chapter 26.

A second alternative to the enterprise DCF method outlined earlier is to discount pretax cash flows at a pretax hurdle rate (the market-based cost of capital multiplied by 1 plus the marginal tax rate) to determine a pretax value. This method, however, leads to three fundamental inconsistencies. First, the government calculates taxes on profits after depreciation, not on cash flow after capital expenditures. By discounting pretax cash flow at the pretax cost of capital, you implicitly assume capital investments are tax deductible when made, not as they are depreciated. Furthermore, working-capital investments, such as accounts receivable and inventory, are never tax deductible. Selling a product at a profit, rather than holding inventory, is what leads to incremental taxes. By discounting pretax cash flow at the pretax cost of capital, you incorrectly assume that investments in operating working capital are tax deductible. Finally, it can be shown that even when net investment equals depreciation, the result will be downward biased—and the larger the cost of capital, the larger the bias. This bias occurs because the method is only an approximation, not a formal mathematical relationship. Because of these inconsistencies, we recommend against discounting pretax cash flows at a pretax hurdle rate.

ALTERNATIVES TO DISCOUNTED CASH FLOW

To this point, we've focused solely on discounted cash flow models. Two additional valuation techniques are using the multiples of comparable companies and real options.

Multiples

One simple way that investors and executives value companies is to value a company in relation to the value of other companies, akin to the way a real estate agent values a house by comparing it with similar houses that have recently sold. To do this, first calculate how similar companies are valued as a multiple of a relevant metric, such as earnings, invested capital, or an operating metric like barrels of oil reserves. You can then apply that multiple to the company you are valuing. For example, assume the company's NOPAT equals \$100 million and the typical enterprise-value-to-NOPAT multiple for companies in the industry with similar growth and ROIC prospects is 13 times. Multiplying 13 by \$100 million leads to an estimated value of \$1.3 billion.

Multiples can be a great check on your DCF valuation if done properly. Suppose the value estimated by multiples is \$1.3 billion, but your DCF value is \$2.7 billion. This might be a clue that there is something wrong with your DCF valuation model. Alternatively, it could be that the company you are valuing is expected to perform differently than the comparable companies. Finally, it could be that investors have a different outlook for the entire industry than you do (in which case the multiples of all the comparable companies would be out of line with their DCF value). Of course, it could just be that your multiples valuation wasn't performed properly. Because of their broad use and potential for error, we devote Chapter 18 to valuation using multiples.

In a nutshell, to use multiples properly, you need to carefully choose the multiple and the comparable companies. In the case of earnings multiples, we recommend using ratios of enterprise value to NOPAT rather than price to earnings or enterprise value to earnings before interest, taxes, depreciation, and amortization (EBITDA). We also urge you to be careful when choosing the comparable companies. The comparable companies not only should be in the same industry, but also should have similar performance, as measured by ROIC and growth.

Real Options and Replicating Portfolios

In 1997, Robert Merton and Myron Scholes won the Nobel Prize in economics for developing an ingenious method to value derivatives that avoids the need to estimate either cash flows or the cost of capital.¹⁸ Their model relies

¹⁸ Fischer Black would have been named as a third recipient, but the Nobel Prize is not awarded posthumously.

on what today's economists call a "replicating portfolio." They argued that if a portfolio exists of traded securities whose future cash flows perfectly mimic the security you are attempting to value, the portfolio and security must have the same price. This is known as the law of one price. As long as you can find a suitable replicating portfolio, you need not discount future cash flows.

Given the model's power in valuing derivatives like stock options, there have been many recent attempts to translate the concepts of replicating portfolios to corporate valuation. This valuation technique, commonly known as real options, is especially useful in situations of great uncertainty. Unlike those for financial options, however, replicating portfolios for companies and their projects are difficult to create. Therefore, although option-pricing models may teach powerful lessons, today's applications are limited. Chapter 39 covers valuation using options-based models.

SUMMARY

Our exploration of the most common DCF valuation models has put a particular focus on the enterprise DCF model and the economic-profit model. Each model has its own rationale, and each has an important place in corporate valuation. The remaining chapters in Part Two describe a step-by-step approach to valuing a company. These chapters explain the technical details of valuation, including how to reorganize the financial statements, analyze return on invested capital and revenue growth, forecast free cash flow, compute the cost of capital, and estimate an appropriate terminal value.

REVIEW QUESTIONS

- Exhibit 10.19 displays the income statement and reorganized balance sheet for BrandCo, a consumer products company. Using the methodology outlined in Exhibit 10.7, determine NOPAT for years 1–6. Assume an operating tax rate of 30 percent. Using the methodology outlined in Exhibit 10.10, determine free cash flow for years 1–6.

EXHIBIT 10.19 **BrandCo: Income Statement and Reorganized Balance Sheet**

\$ million							
Income statement	Today	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Revenues	3,777.1	4,041.5	4,304.2	4,583.9	4,859.0	5,126.2	5,382.5
Operating costs	(3,245.1)	(3,435.2)	(3,658.5)	(3,896.3)	(4,130.1)	(4,357.3)	(4,575.1)
Depreciation	(82.9)	(97.0)	(103.3)	(110.0)	(116.6)	(123.0)	(129.2)
Operating profits	449.1	509.2	542.3	577.6	612.2	645.9	678.2
Interest	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)
Earnings before taxes	435.1	495.2	528.3	563.5	598.2	631.9	664.2
Taxes	(130.5)	(148.6)	(158.5)	(169.1)	(179.5)	(189.6)	(199.2)
Net income	304.6	346.6	369.8	394.5	418.7	442.3	464.9
Reorganized balance sheet	Today	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Operating working capital ¹	188.9	202.1	215.2	229.2	242.9	256.3	269.1
Property and equipment	1,510.8	1,616.6	1,721.7	1,833.6	1,943.6	2,050.5	2,153.0
Invested capital	1,699.7	1,818.7	1,936.9	2,062.8	2,186.5	2,306.8	2,422.1
Debt	280.5	280.5	280.5	280.5	280.5	280.5	280.5
Shareholders' equity	1,419.2	1,538.2	1,656.4	1,782.3	1,906.0	2,026.3	2,141.6
Invested capital	1,699.7	1,818.7	1,936.9	2,062.8	2,186.5	2,306.8	2,422.1

¹Accounts payable has been netted against inventory to determine operating working capital.

- BrandCo currently has 65.6 million shares outstanding. If BrandCo's shares are trading at \$57 per share, what is the company's market capitalization (value of equity)? Assuming the market value of debt equals today's book value of debt, what percentage of the company's value is attributable to debt, and what percentage is attributable to equity? When would the market value of debt not equal the book value? Using these weights, compute the weighted average cost of capital. Assume the pretax cost of debt is 8 percent, the cost of equity is 12 percent, and the marginal tax rate is 30 percent.
- Using the next five years of free cash flow computed in Question 1, an estimated continuing value at the end of year 5, and the weighted average cost of capital computed in Question 2, estimate BrandCo's enterprise value. Assume a long-term growth rate in cash flows of 5 percent and a RONIC of 15 percent. (BrandCo currently has no nonoperating assets.)

4. Assuming the market value of debt equals today's book value of debt, what is the intrinsic equity value for BrandCo? What is the value per share? Does it differ from the share price used to determine the cost-of-capital weightings in Question 2?
5. What are the three components required to calculate economic profit? Determine BrandCo's economic profit in years 1–6.
6. Using the economic profit for years 1–5 calculated in Question 5, an estimated continuing value at the end of year 5, and the weighted average cost of capital computed in Question 2, value BrandCo using the economic-profit-based key value driver model. Assume a long-term growth rate in cash flows of 5 percent and a RONIC of 15 percent. Should discounted economic profit be greater than, equal to, or less than discounted free cash flow? Hint: Prior-year invested capital must be used to determine ROIC and capital charge.
7. Using the information from Exhibit 10.19 and calculations from Question 1, calculate BrandCo's equity value using the cash-flow-to-equity model. How does your valuation compare with your answers in Questions 4 and 6?
8. Exhibit 10.20 displays key financial information for VidCo, an entertainment firm. Fill in the shaded areas of the exhibit to arrive at a value of the operations using the adjusted-present-value model. Assume that the interest tax shields are as risky as the overall firm.

EXHIBIT 10.20 **VidCo: Key Financial Figures**

\$ million

Forecast year	FCF	Discounted FCF	Prior-year debt	Interest rate, %	Interest payment	Marginal tax rate, %	Interest tax shield (ITS)	Discounted ITS
1	100.0		1,000.0	8.0		35.0		
2	105.0		1,040.0	8.0		35.0		
3	110.3		1,081.6	8.0		35.0		
4	115.8		1,124.9	8.0		35.0		
5	121.6		1,169.9	8.0		35.0		
6	127.6		1,216.7	8.0		35.0		
7	134.0		1,265.3	8.0		35.0		
8	140.7		1,315.9	8.0		35.0		
9	147.7		1,368.6	8.0		35.0		
10	155.1		1,423.3	8.0		35.0		
Continuing value			1,480.2	8.0		Continuing value		
Sum of discounted FCFs						Sum of discounted ITSs		
Marginal tax rate, %	35.00							
Unlevered cost of equity, %	10.00							
						PV of FCF and ITS		
						Midyear adjustment factor		
						Value of operations		